

Control of a string of carts

Consider the problem of controlling the string of five carts depicted in the following figure.



where a force u_i can be exerted on each cart, and where the corresponding parameter values are specified in the corresponding MATLAB file. The mathematical model related to the position p_i of generic i -th cart (i.e., the displacement of its position with respect to the rest position) is

$$m_i \ddot{p}_i = u_i + k_{i,i+1}(p_{i+1} - p_i) + k_{i,i-1}(p_{i-1} - p_i) + h_{i,i+1}(\dot{p}_{i+1} - \dot{p}_i) + h_{i,i-1}(\dot{p}_{i-1} - \dot{p}_i)$$

Define, for all $i=1, \dots, 5$, $\dot{p}_i = v_i$, and

$$x = \begin{bmatrix} p_1 \\ v_1 \\ \vdots \\ p_5 \\ v_5 \end{bmatrix}, u = \begin{bmatrix} u_1 \\ \vdots \\ u_5 \end{bmatrix}$$

The system “centralized” model $\dot{x} = Ax + Bu$ is given in the corresponding MATLAB file. We assume that all the state variables are measurable.

Problem:

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time selected compatibly with the continuous-time dynamics). Perform the following analysis:
 - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
 - b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
 - a. Compute the continuous-time fixed modes
 - b. Compute the discrete-time fixed modes
 - c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
 - e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop trajectories (generated both in

continuous-time and in discrete-time) of the cart positions p_i starting from a common random initial condition.